Jayraj Kiran Ladkat

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Summary

I build agentic systems and quantize deep learning models. At TIGER-Lab (University of Waterloo) I am building an iterative harness that improves how open- and closed-source models write code for rendered video, judged by a vision-language model (VLM). I co-authored a KL-guided mixed-precision quantization-aware training method for sub-4-bit inference (IEEE HiPC 2026). I use VLMs to evaluate generated video and am building up core computer vision (VLMs, segmentation) for deploying quantized models on edge devices.

Education

10/2022 – 10/2026 **B.Tech. in Computer Science and Engineering** MIT World Peace University
Pune, Maharashtra, India

Research Interests

Agentic Systems and Harness Optimisation, Model Quantization and Efficient Inference, LLM/VLM Evaluation, and AI for Education

Research Experience

Research Intern 07/2026 – Present
TIGER-Lab, University of Waterloo *Waterloo, Canada (Remote)*

- Advisor: Max Ku.
- Building an iterative harness-improvement loop for code-based video generation (Manim, HyperFrames) across open- and closed-source models, improving on the pi agent harness baseline. Three agents divide the work: a *creator* writes the render code, an *explorer* probes the harness, and an *evaluator* scores the result.
- The explorer runs a genetic algorithm over combinations of prompts and user-preference representations, selecting for the widest diversity of behaviour changes at each step. It tests how well the harness follows a 3-tier, 14-dimension preference taxonomy (content, delivery, visual style).
- Cut context bloat by 35% through progressive disclosure, loading render-API documentation and tools only when the agent needs them.
- Evaluates rendered videos with a persona-conditioned VLM-as-judge. Over the pi agent baseline: +11% render success rate and a +4 judge score.

Quantization Research *IEEE HiPC 2026*
MIT World Peace University *Pune, Maharashtra*

- Co-authored a co-adaptive mixed-precision QAT method: a budgeted controller computes a per-layer marginal KL-divergence signal *during* training and re-allocates weight precision (2/4/8-bit, per-channel STE fake quantization, int8 activations) while the weights are still learning.
- At a 3-bit average budget it beats a post-hoc KL allocator by about 3× in perplexity on Mamba-130M (314 vs. 950), and the advantage transfers to GPT-2 (56.4 vs. 145.1). The gain holds only below a 4-bit average; at 4 bits and above, uniform quantization wins.

Publications

P2. Shreya Mandekar, **Jayraj Kiran Ladkat**, Madhura Phatak, and Bhavana Tiple, "Co-Adaptation over Allocation: KL-Guided Mixed-Precision Quantization-Aware Training," *accepted poster*, IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC), 2026. [Acceptance Proof]

P1. **Jayraj Kiran Ladkat** and Pranati Waghodekar, "Real-World Scalability of PostgreSQL: Practical Techniques Use Case Study," in *IEEE International Conference on Blockchain and Distributed Systems Security* (ICBDS 2025). [IEEE Xplore 11377486]

Professional Experience

Engineering Intern 08/2025 – 12/2025
Publicis Sapient *Gurugram, Haryana*

- Optimised an AI agent data-generation pipeline: reduced per-run token usage, redesigned the help interface, and improved synthetic dataset quality.
- Extended an AI recruitment screening tool with request prefetching, a responsive multi-device frontend, and structured assessment logic, cutting evaluation turnaround and manual HR review effort.

Software Engineering Intern 06/2025 – 08/2025
Elixir AI *Pune, Maharashtra*

- Built the full-stack geospatial application (Flask, PostgreSQL/PostGIS, React, Docker) for the *Climate Smart Public Health Platform*, a Harvard University–Malagasy Ministry of Public Health collaboration.

Cloud & Full-Stack Intern 01/2025 – 02/2025
Centre for Development of Advanced Computing (C-DAC) *Pune, Maharashtra*

- Introduced NATS-based asynchronous messaging into the service architecture, reducing request wait time under load.

Selected Projects

Dexterio — Autonomous Educational Animation Generation Engine 2026

- Engineered an agentic RAG pipeline orchestrating multi-step lesson storyboard planning and modular Manim animation code generation grounded in course materials.
- Architected a distributed worker fleet using Apache Kafka, Daytona/Docker remote VM sandboxes for safe code compilation, and PostgreSQL-backed LangGraph checkpointers for stateful thread resumption.

StudEx — RAG Tutors for Core CS Subjects 2024 – 2025

- Built an educational platform with subject-specific chatbots (Theory of Computation, ML, Blockchain, Cloud) grounded in course PDFs via OpenAI embeddings and a Pinecone vector store (LangChain orchestration).

Technical Skills

Agentic systems:	Multi-agent harnesses, LangGraph, genetic/evolutionary prompt search, LLM/VLM-as-judge evaluation, progressive disclosure, agentic RAG (LangChain, Pinecone)
Quantization:	Quantization-aware training, mixed-precision allocation, STE fake quantization, KL-based sensitivity analysis, PyTorch
Vision:	VLM-based evaluation of rendered video
Languages:	Python, TypeScript, JavaScript, SQL, C/C++
Systems:	Docker, Apache Kafka, NATS, PostgreSQL/PostGIS, AWS (ECS, S3), Flask, Node.js, React, Next.js, Git